

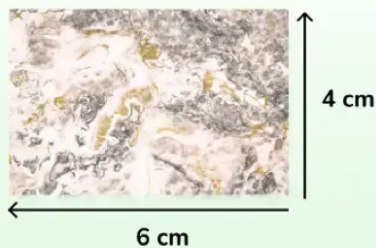
LCM - Least Common Multiple

The Least Common Multiple (LCM) of two numbers is the Smallest Positive number that is evenly divisible by both of them.

For Example, the LCM of 4 and 6 is 12, as 12 is the Smallest number that is divisible by both 4 and 6

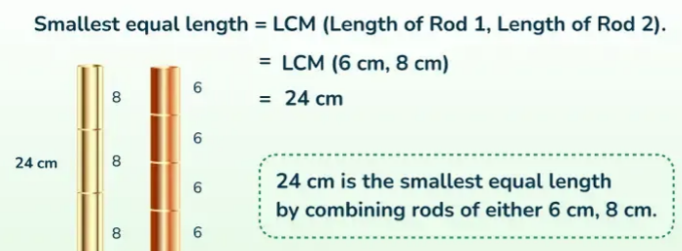
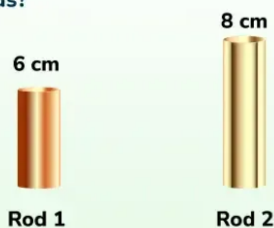
Example 1

What is the smallest side length of a square that can be formed by combining the given tiles?



Example 2

What is the smallest equal length formed by combining following two rods?



Example of LCM (Least Common Multiple)

- The LCM is always greater than or Equal to the larger of the two numbers. The LCM is Equal to the larger of the two numbers when one of the numbers is divisible by the other.
For Example the LCM of 4 and 8 is 8
Because 8 is divisible by 4

- The LCM of two different numbers is multiplication of them if there are no common factors other than 1.
For Example, LCM of 3 and 7 is 21.
- The LCM of two same numbers is the number itself.
For Example LCM of 5 and 5 is 5

LCM of 4 and 6 :

Multiples of 4 are 4, 8, 12, 16, 20.....

Multiples of 6 are 6, 12, 18, 24, 30.....

The lowest common multiple of 4 and 6 is 12.

LCM of 12 and 18:

Multiples of 12 are 12, 24, 36, 48, 60

Multiples of 18 are 18, 36, 54, 72

The lowest common multiple of 12 and 18 is 36.

LCM of 20 and 15:

Multiples of 15 are 15, 30, 45, 60

Multiples of 20 are 20, 40, 60, 80

The lowest common multiple of 20 and 15 is 60.

Approach to find LCM of Two numbers

Approach 1 : Using Conditional loop [Naive Approach]

This Approach to calculating the Least Common Multiple (LCM) involves starting from the greater of the two numbers and checking if it is divisible by the smaller number.

It iterates through multiples of the larger number. Incrementing by the larger number itself in each step.

The first multiple that is divisible by the smaller number is the LCM.

NOTE : This Approach can be not efficient for larger numbers

```
static int lcm(int a, int b) {
```

```
    // Larger value
```

```
    int g = Math.max(a, b);
```

finding max

```
// Smaller value
int s = Math.min(a, b);
```

```
for (int i = g; i <= a * b; i += g) {
    if (i % s == 0)
        return i;
}
```

```
return a * b;
```

```
}
```

finding small

"i" is incremented in multiple of "g"

Loop

if "s" divides "i" then it's a LCM of a, b

Approach 2

using the HCF LCM formula

Relation between HCF and LCM

$$\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$$

$$\text{HCF}(16, 28) \times \text{LCM}(16, 28) = 16 \times 28$$

$$4 \times 112 = 448$$

↑
HCF

↑
LCM

↑
Product of the Numbers

$$a \times b = \text{LCM}(a, b) \times \text{HCF}(a, b)$$

$$\text{LCM}(a, b) = (a \times b) / \text{HCF}(a, b) \Rightarrow \text{LCM}(a, b) = \frac{a}{\text{HCF}(a, b)} \times b$$

// function for gcd

```
static int gcd(int a, int b) {
    return (b == 0) ? a : gcd(b, a % b);
}
```

```
static int lcm(int a, int b) {
    return (a / gcd(a, b)) * b;
}
```

find the HCF of a, b and substitute in the formula

Just started with Basic Algorithms

Tomorrow I will explore more about this Maths Algorithms